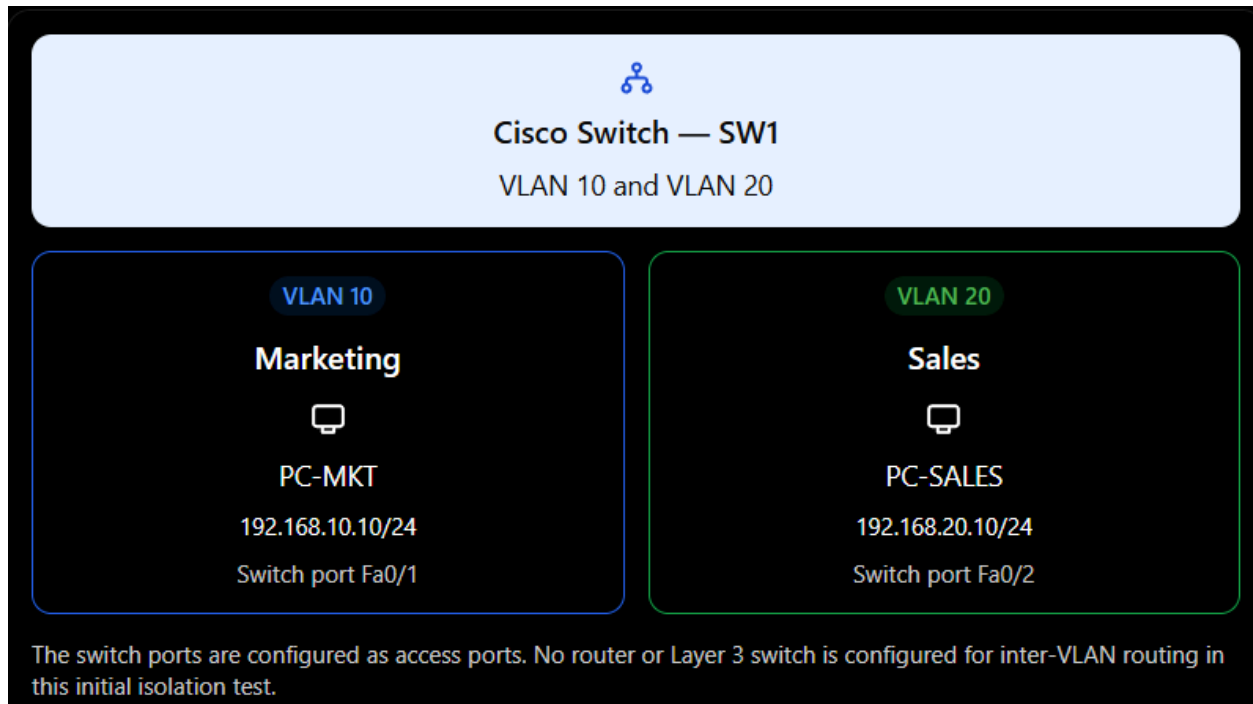


Task 1: Create Two VLANs in a Virtual Lab

Network topology



VLAN configuration commands

Open the switch CLI in Cisco Packet Tracer and enter the following commands:

```
enable
configure terminal
```

```
vlan 10
 name Marketing
exit
```

```
vlan 20
 name Sales
exit
```

```
interface fastEthernet 0/1
```

```
switchport mode access
switchport access vlan 10
no shutdown
exit
```

```
interface fastEthernet 0/2
switchport mode access
switchport access vlan 20
no shutdown
exit
```

```
end
show vlan brief
```

Task 2: Configure IP Subnets and Verify VLAN Isolation

A. IP addressing table

Device	VLAN	IP address	Subnet mask
PC-MKT	10	192.168.10.10	255.255.255.0
PC-SALES	20	192.168.20.10	255.255.255.0

The two networks are:

- Marketing: 192.168.10.0/24
- Sales: 192.168.20.0/24

B. Configure the PCs

In Packet Tracer, select a PC and open Desktop → IP Configuration. Enter the IP address and subnet mask from the table. Leave the default gateway blank for the initial isolation test.

C. Test connectivity

Open Desktop → Command Prompt on each PC.

Test 1: Same VLAN communication

If you add a second Marketing PC, run this command from PC-MKT:

- ping 192.168.10.11

Expected result: Successful replies, provided both PCs are connected and configured correctly.

Test 2: Communication between different VLANs

From PC-MKT, run:

- ping 192.168.20.10

Expected result: The ping fails because the devices belong to different VLANs and no inter-VLAN routing has been configured.

Task 3: Configure a Firewall Rule Between VLANs

Objective

To block all traffic between Marketing and Sales while allowing HTTP traffic (TCP port 80) from Marketing to Sales.

A. Firewall policy

Source	Destination	Protocol / Port	Action
Marketing	Sales	TCP 80 (HTTP)	Allow
Marketing	Sales	All other traffic	Deny
Sales	Marketing	All new connections	Deny

Other networks	Both VLANs	As per policy	Deny unless authorized
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B. Example firewall rule

The following is a simulated firewall policy. It assumes PC-MKT is 192.168.10.10 and the Sales web server is 192.168.20.10.

Rule 1:

Source: 192.168.10.10

Destination: 192.168.20.10

Protocol: TCP

Port: 80

Action: ALLOW

Rule 2:

Source: Marketing subnet

Destination: Sales subnet

Protocol: ANY

Action: DENY

Rule 3:

Source: Sales subnet

Destination: Marketing subnet

Protocol: ANY (new connections)

Action: DENY

C. Testing the firewall policy

For the HTTP test, configure a web server in the Sales VLAN at 192.168.20.10 and enable its HTTP service. Configure routing and firewall policies first.

From the Marketing PC, open the web browser and enter:

- http://192.168.20.10

Expected result: The web page loads.

Then test other traffic:

- ping 192.168.20.10

Expected result: The ping is blocked by the firewall policy.

D. Results table

Test	Expected result
Marketing → Sales, HTTP	Allowed
Marketing → Sales, ICMP ping	Blocked
Marketing → Sales, other ports	Blocked
Sales → Marketing, new connection	Blocked
Sales web server → Marketing, HTTP response	Allowed for the established session

Task 4: Three Real-World Applications of Network Segmentation

Network segmentation can help companies separate systems according to their function and sensitivity. The following are examples of how three well-known services could apply it.

1. Zomato — Food Delivery Platform

Zomato could separate customer-facing application servers, restaurant partner systems, payment services and customer databases into different network segments. If an application server is compromised, firewall rules could prevent the attacker from directly accessing payment systems or customer records.

2. IRCTC — Railway Booking Platform

IRCTC could isolate its public booking website from reservation databases, payment processing systems and internal administrative networks. This would limit unauthorized access to booking records and reduce the spread of an attack between systems.

3. Paytm — Digital Payments Platform

Paytm could place payment processing systems, customer account services, employee workstations and administrative systems in separate security zones. Strict firewall policies could restrict payment-system access to authorized services, reducing exposure if an employee device or another application is compromised.